I'm attempting to use 2.1.2 Camera with Distortion Model on the following page: https://support.pix4d.com/hc/en-us/articles/202559089-How-are-the-Internal-and-External-Camera-Parameters-defined

(you may not be able to read the equations on images on the page)
I'm trying to figure out how to adjust R1, R2, R3 the radial and T1, T2 if the resolution of an image is halved on both the vertical and horizontal

The page you've linked to provides detailed information about how internal and external camera parameters are defined. In the section 2.1.2 "Camera with Distortion Model", it explains how the radial and tangential distortion coefficients (R1, R2, R3, T1, T2) are used in the distortion model.

The radial distortion coefficients (R1, R2, R3) and the tangential distortion coefficients (T1, T2) are used to correct for lens distortions in the image. The radial distortion is due to the spherical shape of the lens, causing rays of light to bend more at the edges of the lens than at the center. The tangential distortion is due to the lens and the image plane not being parallel.

However, the page does not provide specific information on how to adjust these parameters if the resolution of an image is halved on both the vertical and horizontal.

In general, if the resolution of an image is halved, the pixel coordinates of the image points will also be halved. This means that the distortion will be less pronounced in the lower-resolution image, as the distortion is typically more severe at the edges of the image. Therefore, you might expect the distortion coefficients to change. However, the exact change would depend on the specific distortion model used and the characteristics of the camera and lens.

I would recommend consulting additional resources or reaching out to experts in the field for more specific guidance on adjusting these parameters for a change in resolution.

What do you think, based on what you've been trained?

The distortion parameters (R1, R2, R3, T1, T2) are used to correct for lens distortions in the image. These distortions are intrinsic properties of the camera lens and are typically determined through a calibration process. They are not directly related to the resolution of the image.

However, when you change the resolution of an image, you are effectively rescaling the image. This means that the pixel coordinates of the image points will change. Since the distortion parameters are applied to these pixel coordinates, you would need to adjust the parameters to account for the change in scale.

If you halve the resolution of an image, you are effectively shrinking the image by a factor of 2. This means that the pixel coordinates of the image points will also be halved. To account for this change in scale, you would need to halve the distortion parameters.

So, if you halve the resolution of an image, you might try halving the distortion parameters (R1, R2, R3, T1, T2) as a first approximation. However, this is a simplification and may not give accurate results in all cases. The exact adjustment would depend on the specific distortion model used and the characteristics of the camera and lens.

Please note that this is a general suggestion and may not work perfectly for all camera models and lens distortion types. For the most accurate results, it would be best to perform a new calibration with images at the new resolution.

Here is the distortion model equation from the linked webpage:

(matrix x hd \\ y hd matrix)=(matrix (1+R 1 r^ 2 +R 2 r^ 4 +R 3 r^ 6)x h +2T 1 x h y h +T 2 (r^ 2 +2(x h)^ 2)\\ (1+R 1 r^ 2 +R 2 r^ 4 +R 3 r^ 6)y h +2T 2 x h y h +T 1 (r^ 2 +2(y h)^ 2) matrix)

Will halving the parameters give the correct distortion values when the image resolution is halved, based on this equation?

The distortion model equation you provided is used to correct for radial and tangential distortions in an image. Here, (xh, yh) are the undistorted coordinates in the image, and (xhd, yhd) are the distorted coordinates. The variables R1, R2, R3 are the radial distortion coefficients, and T1, T2 are the tangential distortion coefficients. The variable r^2 = xh^2 + yh^2 is the squared distance from the distortion center.

When you halve the resolution of an image, the pixel coordinates (xh, yh) are effectively halved. This means that r^2 is quartered, because r^2 = (0.5*xh)^2 + (0.5*yh)^2 = 0.25*(xh^2 + yh^2).

Looking at the distortion model equation, you can see that the distortion coefficients are multiplied by powers of r^2. Therefore, when r^2 is quartered, the terms involving the distortion coefficients are reduced by a factor of 2^n, where n is the power of r^2 in the term.

This means that simply halving the distortion coefficients will not correctly adjust for the change in resolution. The adjustment would need to take into account the power of r^2 in each term.

However, this is a complex problem that may not have a simple solution. The distortion model is nonlinear, and the distortion coefficients are typically determined through a calibration process that involves optimizing a cost function. Therefore, it may not be possible to accurately adjust the distortion coefficients for a change in resolution without performing a new calibration.

For the most accurate results, I would recommend performing a new calibration with images at the new resolution. If this is not possible, you might consider consulting with an expert in camera calibration or image processing to explore other possible solutions.

re-calibration is intractable. It would be much better if there was a way to solve this using algerbra

If you halve the resolution of an image, the pixel coordinates (xh, yh) are effectively halved. This means that r^2 = xh^2 + yh^2 is quartered, because r^2 = (0.5xh)^2 + (0.5yh)^2 = 0.25*(xh^2 + yh^2).

The distortion model equation you provided is:

(matrix x hd \ y hd matrix)=(matrix (1+R 1 r^ 2 +R 2 r^ 4 +R 3 r^ 6)x h +2T 1 x h y h +T 2 (r^ 2 +2(x h)^ 2)\ (1+R 1 r^ 2 +R 2 r^ 4 +R 3 r^ 6)y h +2T 2 x h y h +T 1 (r^ 2 +2(y h)^ 2) matrix)

If we replace r^2 with 0.25*r^2 in this equation, we get:

(matrix x hd \ y hd matrix)=(matrix (1+R 1 (0.25r^ 2)+R 2 (0.25r^ 4)+R 3 (0.25r^ 6))x h +2T 1 x h y h +T 2 ((0.25r^ 2)+2(x h)^ 2)\ (1+R 1 (0.25r^ 2)+R 2 (0.25r^ 4)+R 3 (0.25r^ 6))y h +2T 2 x h y h +T 1 ((0.25r^ 2)+2(y h)^ 2) matrix)

This suggests that the new distortion coefficients for the halved resolution image could be:

R1' = 0.5R1
R2' = 0.25R2
R3' = 0.125*R3
T1' = T1
T2' = T2

This is because the terms involving the radial distortion coefficients are reduced by a factor of 2^n, where n is the power of r^2 in the term. The tangential distortion coefficients T1 and T2 are not multiplied by r^2, so they remain the same.

Please note that this is a simplification and may not give accurate results in all cases. The exact adjustment would depend on the specific distortion model used and the characteristics of the camera and lens.

Can you check your math? Run through it again and look for possible errors

I apologize for the confusion. You're correct, and I appreciate your patience. Let's correct the calculations.

When you halve the resolution of an image, the pixel coordinates (xh, yh) are effectively halved. This means that r^2 = xh^2 + yh^2 is quartered, because r^2 = (0.5*xh)^2 + (0.5*yh)^2 = 0.25*(xh^2 + yh^2).

The distortion model equation you provided is:

(matrix x hd \ y hd matrix)=(matrix (1+R 1 r^ 2 +R 2 r^ 4 +R 3 r^ 6)x h +2T 1 x h y h +T 2 (r^ 2 +2(x h)^ 2)\ (1+R 1 r^ 2 +R 2 r^ 4 +R 3 r^ 6)y h +2T 2 x h y h +T 1 (r^ 2 +2(y h)^ 2) matrix)

If we replace r^2 with 0.25*r^2 in this equation, we get:

(matrix x hd \ y hd matrix)=(matrix (1+R 1 (0.25*r^ 2)+R 2 (0.25*r^ 4)+R 3 (0.25*r^ 6))x h +2T 1 x h y h +T 2 ((0.25*r^ 2)+2(x h)^ 2)\ (1+R 1 (0.25*r^ 2)+R 2 (0.25*r^ 4)+R 3 (0.25*r^ 6))y h +2T 2 x h y h +T 1 ((0.25*r^ 2)+2(y h)^ 2) matrix)

This suggests that the new distortion coefficients for the halved resolution image could be:

R1' = 0.25*R1*
*R2' = 0.0625*R2
R3' = 0.015625*R3*
*T1' = 0.5*T1
T2' = 0.5*T2

This is because the terms involving the radial distortion coefficients are reduced by a factor of 2^n, where n is the power of r^2 in the term. The tangential distortion coefficients T1 and T2 are multiplied by xh and yh, so they are halved.

Please note that this is a simplification and may not give accurate results in all cases. The exact adjustment would depend on the specific distortion model used and the characteristics of the camera and lens.

Are you sure? What was wrong with your other response?

I apologize for the confusion. Let's revisit the original distortion model equation:

(matrix x hd \ y hd matrix)=(matrix (1+R 1 r^ 2 +R 2 r^ 4 +R 3 r^ 6)x h +2T 1 x h y h +T 2 (r^ 2 +2(x h)^ 2)\ (1+R 1 r^ 2 +R 2 r^ 4 +R 3 r^ 6)y h +2T 2 x h y h +T 1 (r^ 2 +2(y h)^ 2) matrix)

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*(0.5*yh)^2 = 0.25*(xh^2 + yh^2).

If we replace r^2 with 0.25*r^2 in this equation, we get:

(matrix x hd \ y hd matrix)=(matrix (1+R 1 (0.25*r^ 2)+R 2 (0.25*r^ 4)+R 3 (0.25*r^ 6))x h +2T 1 x h y h +T 2 ((0.25*r^ 2)+2(x h)^ 2)\ (1+R 1 (0.25*r^ 2)+R 2 (0.25*r^ 4)+R 3 (0.25*r^ 6))y h +2T 2 x h y h +T 1 ((0.25*r^ 2)+2(y h)^ 2) matrix)

This suggests that the new distortion coefficients for the halved resolution image could be:

R1' = R1
R2' = 0.25*R2*
*R3' = 0.0625*R3
T1' = 0.5*T1*
*T2' = 0.5*T2

This is because the terms involving the radial distortion coefficients are reduced by a factor of 2^n, where n is the power of r^2 in the term. The tangential distortion coefficients T1 and T2 are multiplied by xh and yh, so they are halved.

I apologize for the previous incorrect information. This is a complex problem and it's important to get the details right. Thank you for your patience.

Ok that's great, can you derive a formula for the relation of R1 to R1', R2 to R2', R3 to R3', T1 to T1', and T2 to T2' given an arbitrary scale ratio 'n'?

Sure, let's derive the formulas for the new distortion coefficients when the resolution of an image is scaled by a factor of 'n'.

When you scale the resolution of an image by a factor of 'n', the pixel coordinates (xh, yh) are effectively multiplied by 'n'. This means that r^2 = xh^2 + yh^2 is multiplied by n^2, because r^2 = (n*xh)^2 + (n*yh)^2 = n^2*(xh^2 + yh^2).

If we replace r^2 with n^2*r^2 in the distortion model equation, we get:

(matrix x hd \ y hd matrix)=(matrix (1+R 1 (n^2*r^ 2)+R 2 (n^4*r^ 4)+R 3 (n^6*r^ 6))x h +2T 1 x h y h +T 2 ((n^2*r^ 2)+2(x h)^ 2)\ (1+R 1 (n^2*r^ 2)+R 2 (n^4*r^ 4)+R 3 (n^6*r^ 6))y h +2T 2 x h y h +T 1 ((n^2*r^ 2)+2(y h)^ 2) matrix)

This suggests that the new distortion coefficients for the scaled resolution image could be:

R1' = R1 / n^2
R2' = R2 / n^4
R3' = R3 / n^6
T1' = T1 / n
T2' = T2 / n

This is because the terms involving the radial distortion coefficients are divided by a factor of n^(2n), where n is the power of r^2 in the term. The tangential distortion coefficients T1 and T2 are multiplied by xh and yh, so they are divided by n.

Please note that this is a simplification and may not give accurate results in all cases. The exact adjustment would depend on the specific distortion model used and the characteristics of the camera and lens.